

Homework 10

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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December 2025

Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

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Problem 1 (*Ross Chapter 3 18.5 (a)*)

Let f and g be continuous functions in $[a, b]$ such that $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Prove $f(x_0) = g(x_0)$ for at least one x_0 in $[a, b]$.

Discussion.

We want a point where $f(x_0) = g(x_0)$. This is equivalent to $f(x_0) - g(x_0) = 0$, so it is natural to consider the difference $h(x) = f(x) - g(x)$. Since f and g are continuous, h is continuous on $[a, b]$. The inequalities $f(a) \geq g(a)$ and $f(b) \leq g(b)$ then say $h(a) \geq 0$ and $h(b) \leq 0$, so 0 lies between $h(a)$ and $h(b)$. By the Intermediate Value Theorem applied to h , there is some $x_0 \in [a, b]$ with $h(x_0) = 0$, i.e. $f(x_0) = g(x_0)$.

Proof.

Define $h : [a, b] \rightarrow \mathbb{R}$ by

$$h(x) = f(x) - g(x).$$

Since f and g are continuous on $[a, b]$ and the difference of two continuous functions is continuous, it follows that h is continuous on $[a, b]$.

From the hypotheses we have

$$f(a) \geq g(a) \implies h(a) = f(a) - g(a) \geq 0,$$

and

$$f(b) \leq g(b) \implies h(b) = f(b) - g(b) \leq 0.$$

Thus 0 lies between $h(a)$ and $h(b)$; equivalently, $h(b) \leq 0 \leq h(a)$.

By the Intermediate Value Theorem, since h is continuous on $[a, b]$ and 0 lies between $h(a)$ and $h(b)$, there exists some $x_0 \in [a, b]$ such that

$$h(x_0) = 0.$$

By the definition of h this means

$$f(x_0) - g(x_0) = 0,$$

so $f(x_0) = g(x_0)$.

Therefore there exists at least one $x_0 \in [a, b]$ with $f(x_0) = g(x_0)$, as required. $\square$

Problem 2 (*Ross Chapter 3 18.10*)

Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Prove there exist x, y in $[0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$. *Hint:* Consider $g(x) = f(x + 1) - f(x)$ on $[0, 1]$.

Discussion.

We want two points x, y with $|y - x| = 1$ and $f(x) = f(y)$. Writing $y = x + 1$, this is equivalent to finding $x \in [0, 1]$ such that

$$f(x + 1) - f(x) = 0.$$

So it is natural to define

$$g(x) = f(x + 1) - f(x), \text{ for } x \in [0, 1].$$

Because f is continuous on $[0, 2]$, both $f(x + 1)$ and $f(x)$ are continuous on $[0, 1]$, so their difference g is continuous on $[0, 1]$. The condition $f(0) = f(2)$ then forces

$$g(0) = f(1) - f(0), \quad g(1) = f(2) - f(1) = f(0) - f(1) = -g(0),$$

so either $g(0) = 0$ or $g(0)$ and $g(1)$ have opposite signs. In either case, 0 lies between $g(0)$ and $g(1)$, so by the Intermediate Value Theorem there is some $x \in [0, 1]$ with $g(x) = 0$, i.e. $f(x + 1) = f(x)$ and $|(x + 1) - x| = 1$.

TL;DR: We basically have to translate the statement we are trying to prove to make it in terms of just x (which we do by defining g) to then make the question about applying the IVT to g to prove that it has a zero, which then maps back to the original statement we are trying to prove about f .

Proof.

Let g be a function defined on $[0, 1]$ as

$$g(x) = f(x + 1) - f(x).$$

Since f is continuous on $[0, 2]$ (and $[0, 1] \subset [0, 2]$), then g must be continuous on $[0, 1]$.

We compute the endpoint values of g :

$$g(0) = f(1) - f(0),$$

and, using $f(0) = f(2)$,

$$g(1) = f(2) - f(1) = f(0) - f(1) = -g(0).$$

If $g(0) = 0$, then taking $x = 0$ and $y = 1$ gives

$$|y - x| = |1 - 0| = 1, \quad f(x) = f(0) = f(1) = f(y),$$

so the desired property holds.

Otherwise $g(0) \neq 0$, so $g(1) = -g(0)$ has the opposite sign. Thus

$$g(0) \cdot g(1) < 0,$$

which means 0 lies strictly between $g(0)$ and $g(1)$. Since g is continuous on $[0, 1]$, the Intermediate Value Theorem guarantees the existence of some $x \in (0, 1)$ such that

$$g(x) = 0.$$

For this x , set $y = x + 1$. Then $x, y \in [0, 2]$, we have

$$|y - x| = |x + 1 - x| = 1,$$

and

$$f(y) = f(x + 1) = f(x),$$

because $g(x) = f(x + 1) - f(x) = 0$.

Therefore there exist $x, y \in [0, 2]$ with $|y - x| = 1$ and $f(x) = f(y)$, as required. $\square$

Problem 3 (*Ross Chapter 3 19.6 (b)*)

(From Part (a):) Let $f(x) = \sqrt{x}$ for $x \geq 0$. Show f' is unbounded on $(0, 1]$ but f is nevertheless uniformly continuous on $(0, 1]$ Compare with Theorem 19.6.

(b) Show that f is uniformly continuous on $[1, \infty)$.

Discussion.

We want to prove that $f(x) = \sqrt{x}$ is *uniformly* continuous on $[1, \infty)$. By Definition 19.1, this means:

for each $\varepsilon > 0$ there exists $\delta > 0$ such that $x, y \in [1, \infty)$, $|x - y| < \delta \implies |\sqrt{x} - \sqrt{y}| < \varepsilon$.

So the game is: fix an arbitrary $\varepsilon > 0$ and then cook up a δ that depends *only* on ε , not on a particular choice of x or y .

A standard trick for square roots is to use the algebraic identity

$$|\sqrt{x} - \sqrt{y}| = \left| \frac{x - y}{\sqrt{x} + \sqrt{y}} \right|.$$

On our domain $[1, \infty)$ we always have $\sqrt{x} \geq 1$ and $\sqrt{y} \geq 1$, so

$$\sqrt{x} + \sqrt{y} \geq 2.$$

That means we can bound the difference by

$$|\sqrt{x} - \sqrt{y}| \leq \frac{|x - y|}{2}.$$

This is the key estimate: on $[1, \infty)$, the function $\sqrt{x}$ cannot change faster than “slope $\frac{1}{2}$ ” in the sense that

$$|\sqrt{x} - \sqrt{y}| \leq \frac{1}{2} |x - y|.$$

Once we have this, the ε - δ choice is easy: if we make $|x - y|$ smaller than 2ε , then $\frac{1}{2}|x - y|$ will automatically be smaller than ε . So in the proof we will take

$$\delta = 2\varepsilon$$

and then check that this choice satisfies the definition of uniform continuity on $[1, \infty)$.

Proof.

Let $\varepsilon > 0$, and $\delta = 2\varepsilon$. From Definition 19.1 let $x, y \in S$ with $|x - y| < \delta$. Then,

$$|\sqrt{x} - \sqrt{y}| = \left| \frac{x - y}{\sqrt{x} + \sqrt{y}} \right|.$$

Because $x, y \in [1, \infty)$, we have $\sqrt{x} \geq 1$ and $\sqrt{y} \geq 1$, so

$$\sqrt{x} + \sqrt{y} \geq 2.$$

Therefore

$$|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{|x - y|}{2}.$$

Using $|x - y| < \delta = 2\varepsilon$, we obtain

$$|\sqrt{x} - \sqrt{y}| \leq \frac{|x - y|}{2} < \frac{2\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, by Definition 19.1, $f(x) = \sqrt{x}$ is uniformly continuous on $[1, \infty)$. □

Problem 4 (*Ross Chapter 3 19.7*)

(a) Let f be a continuous function on $[0, \infty)$. Prove that if f is uniformly continuous on $[k, \infty)$ for some k , then f is uniformly continuous on $[0, \infty)$.

(b) Use (a) and Exercise 19.6(b) to show that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Discussion.

Part (a) seems like it is asking us to sorta glue together two pieces of information about f . We know that f is continuous on $[0, \infty)$, and that for some fixed $k > 0$, f is uniformly continuous on $[k, \infty)$ (i.e. $k = 1$ as we proved in Problem 3).

To prove that f is uniformly continuous on all of $[0, \infty)$, we want to use Theorem 19.2, and the given hypothesis that f is already uniformly continuous on $[k, \infty)$ in tandem to then be able to conclude that f too is uniformly continuous on $[0, \infty)$.

The key idea/innovation is to split the domain into two overlapping pieces:

$$[0, \infty) = [0, k+1] \cup [k, \infty).$$

Because f is continuous on $[0, \infty)$, it is also (by nature of $[0, k+1] \subseteq [0, \infty)$) continuous on the closed interval $[0, k+1]$. From there we can then say, by Theorem 19.2, f is uniformly continuous on $[0, k+1]$ (which could be like $[0, 1]$). Together with the hypothesis, this gives us uniform continuity on $[0, k+1]$ and $[k, \infty)$.

But because $[0, k+1]$ and $[k, \infty)$ are distinct, we need to be careful and sorta believe that they are distinct deltas, so that together with the hypothesis we get:

- uniform continuity on $[0, k+1]$ with some δ_1 for each ε ;
- uniform continuity on $[k, \infty)$ with some δ_2 for each ε .

The main subtlety (hard part) is: how do we turn these two local pieces into a single δ that works for *all* $x, y \in [0, \infty)$ at once?

So after shedding a good tear, we then arrive at the trick. The trick is to choose a δ that is small enough to force any two points x, y with $|x - y| < \delta$ to lie entirely inside *one* of the two regions where we already know f is uniformly continuous. Notice that if $|x - y| < 1$, then it is impossible for one point to lie at or below k and the other to lie at or above $k+1$ (that would give $|x - y| \geq 1$). So whenever $|x - y| < 1$, either

$$x, y \in [0, k+1] \quad \text{or} \quad x, y \in [k, \infty).$$

Therefore, given $\varepsilon > 0$, we can:

- Get $\delta_1 > 0$ from uniform continuity of f on $[0, k+1]$.
- Get $\delta_2 > 0$ from uniform continuity of f on $[k, \infty)$.

(iii) Set

$$\delta = \min\{1, \delta_1, \delta_2\}.$$

Then for any $x, y \in [0, \infty)$ with $|x - y| < \delta$, we automatically have $|x - y| < 1$, so x and y lie in the same piece: either both in $[0, k + 1]$ or both in $[k, \infty)$. In the first case we can use δ_1 ; in the second we use δ_2 . Either way we get $|f(x) - f(y)| < \varepsilon$, which is exactly the definition of uniform continuity on $[0, \infty)$.

For part (b), we simply apply (a) to the specific function $f(x) = \sqrt{x}$. From Exercise 19.6(b) (Problem 3) we already know that f is uniformly continuous on $[1, \infty)$. Since f is continuous on $[0, \infty)$ and is uniformly continuous on $[1, \infty)$, part (a) with $k = 1$ tells us that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Proof.

(a) Let $\varepsilon > 0$. We want to find a single $\delta > 0$ that works for all $x, y \in [0, \infty)$ in the sense of Definition 19.1.

Because f is continuous on $[0, \infty)$, its restriction to the closed interval $[0, k + 1]$ is continuous. Hence, by Theorem 19.2, f is uniformly continuous on $[0, k + 1]$. Thus there exists $\delta_1 > 0$ such that

$$x, y \in [0, k + 1], |x - y| < \delta_1 \implies |f(x) - f(y)| < \varepsilon.$$

By hypothesis, f is uniformly continuous on $[k, \infty)$, so there exists $\delta_2 > 0$ such that

$$x, y \in [k, \infty), |x - y| < \delta_2 \implies |f(x) - f(y)| < \varepsilon.$$

Now define

$$\delta = \min\{1, \delta_1, \delta_2\}.$$

We claim that this δ works for all of $[0, \infty)$.

Let $x, y \in [0, \infty)$ satisfy $|x - y| < \delta$. Then automatically $|x - y| < 1$, so it is impossible for one of x, y to lie at or below k while the other lies at or above $k + 1$ (that would force $|x - y| \geq 1$). Hence we must be in one of two cases:

- *Case 1:* $x, y \in [0, k + 1]$. Then $|x - y| < \delta \leq \delta_1$, so by the choice of δ_1 we have

$$|f(x) - f(y)| < \varepsilon.$$

- *Case 2:* $x, y \in [k, \infty)$. Then $|x - y| < \delta \leq \delta_2$, so by the choice of δ_2 we have

$$|f(x) - f(y)| < \varepsilon.$$

In either case, whenever $x, y \in [0, \infty)$ satisfy $|x - y| < \delta$, we obtain $|f(x) - f(y)| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this is exactly the statement that f is uniformly continuous on $[0, \infty)$.

(b) Let $f(x) = \sqrt{x}$ on $[0, \infty)$. From Problem 3 (Exercise 19.6(b)) we already know that f is uniformly continuous on $[1, \infty)$, so we can apply part (a) with $k = 1$.

The function $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$ which includes the closed interval $[0, 1]$. In other words, f is continuous on the closed interval $[0, 1]$, and by Theorem 19.2 we conclude that f is uniformly continuous on $[0, 1]$.

We have proven in Problem 3 that f is uniformly continuous on $[1, \infty)$. So f is uniformly continuous on both $[0, 1]$ and $[1, \infty)$. Therefore f is uniformly continuous on $[0, \infty)$. $\square$

Problem 5 (*Uniform Continuity of f*)

Suppose $f : \mathbb{R} \mapsto \mathbb{R}$. Prove that if there exists a real number $M \geq 0$ such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Discussion.

The hypothesis says that there is some fixed constant $M \geq 0$ so that

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every pair } x, y \in \mathbb{R}.$$

To my understanding, this inequality says: whenever I move x and y a certain horizontal distance apart, the vertical change in the function values $f(x)$ and $f(y)$ is at most M times that horizontal distance. So the function is never allowed to change “too fast” (non-linearly/in-a-non-uniform-fashion).

To show that f is uniformly continuous on $\mathbb{R}$, I need to line up with Definition 19.1: given any $\varepsilon > 0$, I must find a $\delta > 0$ so that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

for *all* real numbers x and y at once.

The useful part of the hypothesis is that it already compares $|f(x) - f(y)|$ directly to $|x - y|$:

$$|f(x) - f(y)| \leq M|x - y|.$$

If I can make $M|x - y|$ smaller than ε , then automatically $|f(x) - f(y)|$ will also be smaller than ε . So I want to force

$$M|x - y| < \varepsilon.$$

When $M > 0$, this inequality will hold whenever

$$|x - y| < \frac{\varepsilon}{M}.$$

That suggests a very natural choice:

$$\delta = \frac{\varepsilon}{M}.$$

There is one special case to think about: if $M = 0$, then the inequality becomes

$$|f(x) - f(y)| \leq 0 \cdot |x - y| = 0,$$

so in fact $|f(x) - f(y)| = 0$ for all x, y . That means $f(x)$ is the same number for every x ; in other words, f has to be a constant function. A constant function is automatically uniformly continuous, because $|f(x) - f(y)| = 0$ no matter how we choose δ .

So the plan of the proof is:

- First notice that if $M = 0$, then f is constant and therefore uniformly continuous.

- If $M > 0$, then for a given $\varepsilon > 0$ we set $\delta = \varepsilon/M$ and use the inequality $|f(x) - f(y)| \leq M|x - y|$ to verify that $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$.

This will match the definition of uniform continuity on $\mathbb{R}$.

Proof.

Let $\varepsilon > 0$. Let $M \geq 0$ be such that:

Case 1: $M = 0$. Then for all $x, y \in \mathbb{R}$ we have

$$|f(x) - f(y)| \leq 0 \cdot |x - y| = 0,$$

so $|f(x) - f(y)| = 0$ and hence $f(x) = f(y)$ for all x, y . Thus f is constant, and for any choice of $\delta > 0$ and any $x, y \in \mathbb{R}$ with $|x - y| < \delta$ we have

$$|f(x) - f(y)| = 0 < \varepsilon.$$

So f is uniformly continuous on $\mathbb{R}$.

Case 2: $M > 0$. Set

$$\delta = \frac{\varepsilon}{M}.$$

Let $x, y \in \mathbb{R}$ satisfy $|x - y| < \delta$. Then, using the hypothesis,

$$|f(x) - f(y)| \leq M|x - y| < M\delta = M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this shows that for each $\varepsilon > 0$ there exists $\delta > 0$ (namely $\delta = \varepsilon/M$) such that

$$x, y \in \mathbb{R}, |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

By Definition 19.1, f is uniformly continuous on $\mathbb{R}$. □

Problem 6 (*Differentiability of f*)

Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \mapsto \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g .

Hint: Note that you cannot use the product rule, since $g(x)$ is not guaranteed to be differentiable at $x = 0$

Discussion.

To show that $f(x) = xg(x)$ is differentiable at $x = 0$, we use the definition of the derivative (Definition 28.1) directly. Like the hint says, we can't use the product rule because g is only assumed to be continuous at 0, not differentiable (the converse of Theorem 28.2 is not true; differentiability $\implies$ continuity, but not the other way around). So we apply Definition 28.1:

$$\frac{f(x) - f(0)}{x - 0}.$$

Since $f(0) = 0 \cdot g(0) = 0$, this becomes

$$\frac{xg(x) - 0}{x} = g(x) \quad (x \neq 0).$$

So the entire derivative computation reduces to studying the limit $\lim_{x \rightarrow 0} g(x)$. Because g is continuous at 0, we know that

$$\lim_{x \rightarrow 0} g(x) = g(0).$$

Thus the derivative exists and equals $g(0)$. The key idea is that continuity of g gives us that the limit exists and thus f is differentiable directly from the definition of a derivative, which simplifies in a way that avoids product rule entirely. winning

Proof.

Since $f(0) = 0 \cdot g(0) = 0$, for $x \neq 0$ we compute

$$\frac{f(x) - f(0)}{x - 0} = \frac{xg(x)}{x} = g(x).$$

Taking limits,

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} g(x) = g(0),$$

where the last equality follows from continuity of g at 0. Therefore f is differentiable at 0, and

$$f'(0) = g(0).$$

□

Appendix: Theorems

Misc theorems potentially relevant to the homework problems that I thought I might use.

Theorem 17.2

Let f be a real-valued function whose domain is a subset of $\mathbb{R}$. Then f is continuous at $x_0 \in \text{dom}(f)$ if and only if

for each $\varepsilon > 0$ there exists $\delta > 0$ such that $x \in \text{dom}(f)$ and

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Theorem 18.1 — Extreme Value Theorem

Let f be a continuous real-valued function on a closed interval $[a, b]$. Then f is a bounded function. Moreover, f assumes its maximum and minimum values on $[a, b]$; that is, there exist x_0, y_0 in $[a, b]$ such that

$$f(x_0) \leq f(x) \leq f(y_0) \quad \text{for all } x \in [a, b].$$

Theorem 18.2 — Intermediate Value Theorem

If f is a continuous real-valued function on an interval I , then f has the intermediate value property on I : Whenever $a, b \in I$, $a < b$, and y lies between $f(a)$ and $f(b)$ (i.e., $f(a) < y < f(b)$ or $f(b) < y < f(a)$), there exists at least one x in (a, b) such that

$$f(x) = y.$$

Corollary 18.3

If f is a continuous real-valued function on an interval I , then the set

$$f(I) = \{f(x) : x \in I\}$$

is also an interval or a single point.

Definition 19.1 — Uniform Continuity

Let f be a real-valued function defined on a set $S \subseteq \mathbb{R}$. Then f is *uniformly continuous* on S if

for each $\varepsilon > 0$ there exists $\delta > 0$ such that $x, y \in S$ and $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

We will say f is *uniformly continuous* if f is uniformly continuous on $\text{dom}(f)$.

Theorem 19.2

If f is continuous on a closed interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Discussion 19.3

Example 5 illustrates the power of Theorem , but it still may not be clear why uniform continuity is worth studying. One of the important applications of uniform continuity concerns the integrability of continuous functions on closed intervals.

To see the relevance of uniform continuity, consider a continuous nonnegative real-valued function f on $[0, 1]$. For $n \in \mathbb{N}$ and $i = 0, 1, 2, \dots, n-1$, let

$$M_{i,n} = \sup\{f(x) : x \in [\frac{i}{n}, \frac{i+1}{n}]\}, \quad m_{i,n} = \inf\{f(x) : x \in [\frac{i}{n}, \frac{i+1}{n}]\}.$$

Then the upper sum and lower sum are given by

$$U_n = \frac{1}{n} \sum_{i=0}^{n-1} M_{i,n}, \quad L_n = \frac{1}{n} \sum_{i=0}^{n-1} m_{i,n}.$$

The function f would be Riemann integrable provided $U_n - L_n \rightarrow 0$, i.e.,

$$\lim_{n \rightarrow \infty} (U_n - L_n) = 0.$$

Since f is uniformly continuous on $[0, 1]$ by Theorem , there exists $\delta > 0$ such that

$$x, y \in [0, 1], |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Choosing N such that $\frac{1}{N} < \delta$ and $n > N$ gives

$$0 \leq U_n - L_n = \frac{1}{n} \sum_{i=0}^{n-1} (M_{i,n} - m_{i,n}) < \varepsilon.$$

This proves the claim. The next two theorems show uniformly continuous functions have nice properties.

Definition 28.1 — Differentiability

Let f be a real-valued function defined on an open interval that contains a point a . We say that f is *differentiable at a* , or that f *has a derivative at a* , if the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists (as a finite real number).

Theorem 28.2

If f is differentiable at a point a , then f is continuous at a .

Theorem 28.3 — Algebra Rules

Let f and g be functions that are differentiable at a point a , and let c be a real constant. Then each of the following functions is differentiable at a (whenever it is defined), and their derivatives are given by:

(a) **Constant multiple:**

$$(cf)'(a) = c \cdot f'(a).$$

(b) **Sum:**

$$(f + g)'(a) = f'(a) + g'(a).$$

(c) **Product rule:**

$$(fg)'(a) = f(a)g'(a) + f'(a)g(a).$$

(d) **Quotient rule:** If $g(a) \neq 0$, then f/g is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{[g(a)]^2}.$$

Theorem 28.4 — Chain Rule

Assume f is differentiable at a and g is differentiable at $f(a)$. Then the composite function $g \circ f$ is differentiable at a , and

$$(g \circ f)'(a) = g'(f(a)) f'(a).$$